

ANKENY, IA, USA

WMO: 725466

Lat: 41.691N

Lon: 93.566W

Elev: 277

StdP: 98.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04938

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-17.5	-25.2	0.4	-19.2	-22.5	0.5	-16.4	11.9	-4.5	11.0	-4.0	3.4	310	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.2	(c) 33.8	(d) 24.2	(e) 32.4	(f) 24.0	(g) 31.0	(h) 23.3	(i) 26.4	(j) 31.3	(k) 25.3	(l) 30.4	(m) 24.3	(n) 29.1	(o) 4.6	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.8	29.6	23.7	19.2	28.6	22.7	18.0	27.5	84.2	31.2	78.8	30.3	74.6	29.2	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.4	(b) 9.0	(c) 8.1	(e) -24.9	(f) 36.6	(g) 4.0	(h) 2.0	(i) -27.7	(j) 38.0	(k) -30.1	(l) 39.2	(m) -32.3	(n) 40.3	(o) -35.2	(p) 41.8		
(a) 10.4	(b) 9.0	(c) 8.1	(e) -24.8	(f) 27.9	(g) 4.1	(h) 1.1	(i) -27.8	(j) 28.7	(k) -30.1	(l) 29.4	(m) -32.4	(n) 30.0	(o) -35.4	(p) 30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-5.2	-3.1	3.8	10.7	16.6	21.8	23.9	22.7	18.5	11.2	4.4	-2.1
	DBStd	11.44	7.09	7.12	6.68	5.56	4.73	3.42	3.20	3.28	4.45	5.04	5.55	5.99
	HDD10.0	1717	471	369	212	56	4	0	0	0	2	47	181	376
	HDD18.3	3451	729	601	451	236	91	8	1	4	52	226	418	634
	CDD10.0	1839	0	1	21	76	208	355	432	393	256	85	13	1
	CDD18.3	532	0	0	1	6	37	113	174	139	56	6	0	0
	CDH23.3	5266	0	0	13	89	355	1057	1809	1306	561	73	2	0
	CDH26.7	1919	0	0	2	21	101	367	742	491	182	14	0	0
Wind	WSAvg	3.5	4.0	4.0	4.2	4.4	3.7	3.1	2.6	2.4	2.8	3.5	3.8	3.7
Precipitation	PrecAvg	892	23	30	54	104	131	130	109	100	73	59	47	33
	PrecMax	1379	56	52	161	197	245	327	277	291	141	174	126	123
	PrecMin	584	11	6	0	43	39	24	26	1	21	6	3	0
	PrecStd	193	10	13	36	36	53	79	64	72	38	41	29	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	17.6	25.1	29.8	32.5	34.0	37.0	35.9	32.9	28.8	22.7	14.6
		MCWB	7.8	9.7	15.3	19.7	22.0	25.1	23.9	24.3	22.5	19.2	15.1	10.0
	2%	DB	9.9	13.3	21.2	26.0	29.1	32.3	34.1	32.9	31.1	25.2	18.1	11.2
		MCWB	5.4	8.4	13.9	16.5	20.8	24.1	24.8	23.9	22.7	17.9	13.1	7.9
	5%	DB	7.0	9.8	17.5	22.8	27.2	30.5	32.5	31.3	28.8	22.6	16.0	8.0
		MCWB	3.7	6.0	12.4	14.8	18.9	23.1	24.7	23.4	21.6	16.4	11.1	5.3
	10%	DB	3.8	6.4	14.1	20.9	25.0	28.8	30.9	29.7	27.1	20.0	12.9	6.2
		MCWB	1.7	3.5	9.0	13.4	17.8	22.3	24.1	23.3	20.0	14.8	9.0	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	11.4	17.7	20.5	23.9	26.8	28.7	27.1	25.0	22.3	16.4	11.8
		MCDB	12.8	16.3	22.9	26.9	30.0	31.6	32.9	32.5	30.6	26.1	20.6	13.1
	2%	WB	5.6	8.7	15.3	17.9	22.2	25.5	26.9	25.7	23.8	19.8	14.3	8.1
		MCDB	9.3	13.6	19.2	23.5	27.6	30.6	31.3	30.8	28.8	22.9	17.9	9.7
	5%	WB	3.8	5.8	12.4	16.1	20.7	24.3	25.7	24.7	22.6	17.8	11.7	5.7
		MCDB	6.5	9.5	17.3	22.1	25.3	28.8	30.4	29.7	27.2	20.9	15.2	8.5
	10%	WB	2.0	3.7	9.8	14.4	19.4	23.1	24.5	23.6	21.3	15.3	9.6	3.8
		MCDB	4.1	6.4	14.3	19.1	23.4	27.6	29.2	28.2	25.2	19.0	12.8	6.1
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.8	10.7	11.9	11.1	10.7	11.2	11.4	12.5	11.5	10.5	8.6
		MCDBR	13.3	14.1	16.0	16.1	13.6	12.2	12.9	13.4	14.2	15.4	14.9	12.1
		MCWBR	9.7	9.5	9.7	9.3	6.7	6.3	5.9	6.2	7.0	8.6	9.9	9.1
	5% WB	MCDBR	12.5	13.4	14.6	14.1	11.8	10.9	11.1	11.4	11.6	12.0	13.2	11.1
MCWBR		9.5	9.5	9.5	8.9	6.5	6.9	6.3	6.4	6.4	8.3	9.6	8.9	
Clear-Sky Solar Irradiance	taub	0.291	0.317	0.330	0.387	0.415	0.426	0.443	0.425	0.390	0.345	0.301	0.292	
	taud	2.414	2.341	2.401	2.262	2.254	2.270	2.207	2.271	2.319	2.441	2.501	2.429	
	Ebn at Noon	861	887	918	882	863	851	832	836	841	846	846	828	
	Edh at Noon	79	99	105	130	135	133	140	128	114	89	72	72	
All-Sky Solar Radiation	RadAvg	1.92	2.92	3.79	4.75	5.60	6.23	6.30	5.45	4.57	3.07	2.08	1.60	
	RadStd	0.18	0.20	0.36	0.37	0.44	0.59	0.46	0.36	0.34	0.36	0.17	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page